

Automatic high-density polymer waveguide layout for on-board high-speed optical interconnect

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Abstract—We demonstrate automatic polymer waveguide layout for the first time. We succeeded in 512 multimode waveguides routing with balanced performances within an area of $150 \times 150 \text{ mm}^2$ by optimization on algorithm and bending radius.

Keywords—optical interconnect, polymer waveguide, automatic routing

I. INTRODUCTION

The increasing operation speed of datacenters and high performance computers gives ever-increased burden on the bandwidth of intra-system interconnects. Board-level optical interconnects have drawn a lot of attention due to their distinctive advantage in bandwidth over their electric counterparts and are regarded as the potential solutions to meet future data transmission performance requirements. Various methods have been proposed to realize board-level optical interconnects such as free-space optics, optical fiber wiring and planar waveguide technology [1]. Among them, multimode polymer waveguides have been intensively studied due to their good compatibility with printed circuit boards (PCBs) and the excellent tolerance for misalignment [2]. Multimode polymer waveguides combined with 850 nm vertical cavity surface emitting lasers (VCSELs) and photodetectors (PDs) are considered to be the near-term solution for board-level optical interconnects. Meter-scale multimode waveguides with low transmission loss (0.046 dB/cm at a wavelength of 850 nm), low inter-channel crosstalk (-58 dB), large misalignment tolerance ($\pm 20 \text{ }\mu\text{m}$) and good environmental stability has been demonstrated. They also show good performance for 30 Gb/s NRZ and 56 Gb/s PAM4 high-speed data transmission [3,4].

However, both data rate and density for board-level optical interconnects are still continuously increasing and the number of interconnects of a single board may reach hundreds or even thousands of channels [5]. Traditional manual routing method is error-prone, low efficient, and difficult to deal with batch arrangement. PCB layouts are mainly implemented automatically by computer [6-8], but there are few investigations on automatic optical waveguide layout method.

In this paper, we propose a fully automatic multimode polymer waveguides layout method for the first time to the best of our knowledge based on a channel-routing algorithm [9]. The objective is to realize automatic layout of numerous waveguides with fabricability. We analyzed the effects of length, the number of crossings and waveguide bending radius on the overall performance of optical waveguides. We demonstrate automatic layout of high-density multimode polymer waveguides of 512 channels in an area as small as $150 \times 150 \text{ mm}^2$. The performances of the waveguides were balanced and the estimated average channel loss is 9.90 dB

TABLE I. WAVEGUIDE PARAMETERS

Parameter	Value
Wavelength	850 nm
Core index	1.532
Cladding index	1.512
Waveguide width	50 μm
Waveguide height	50 μm
Pitch	125 μm
Loss for straight waveguide	0.05 dB/cm
Additional loss of 90-degree crossing	0.018 dB
Loss of 90-degree bending structure	7.94 dB / 2 mm
	6.81 dB / 3 mm
	4.59 dB / 4 mm
	2.39 dB / 5 mm
	1.90 dB / 6 mm

and 5.61 dB when the bending radius is 4 mm and 5 mm, respectively. Moreover, the proposed method can be easily applied to fiber routing processes for optical backplane applications.

II. MODELING AND ANALYSIS

The main goal of the routing works is to obtain a layout aiming at loss minimization. For the layout design to be considered fabricable, three principles must be followed when routing a large number of polymer waveguides within a limited area: 1) keeping the waveguide pitch; 2) avoiding multiple crossing, where three or more waveguides cross over at a tiny area; 3) increasing the crossing angle and perpendicular crossing are desirable.

In this paper, we demonstrate 512 channels polymer waveguide routing in an area of $150 \times 150 \text{ mm}^2$, where the polymer mechanical transfer (PMT) connectors are arranged along only the top and bottom edges by assuming that the PMTs are set parallelly at the top and the bottom of the footprint and each of the PMT connectors equipped with 16 ports. The interconnection of these PMTs is reflected by an undirected graph set up in advance, where a vertex represents a PMT and an edge represents a waveguide channel. The main parameters adopted for modeling the routing procedure are listed in TABLE I.

Since the PMTs are arranged along only the top and bottom edges, each channel can be split into two vertical lines, one horizontal line and two bends. The problem of waveguides layout can be simplified to determine the two endpoints, the distance of the horizontal line from the

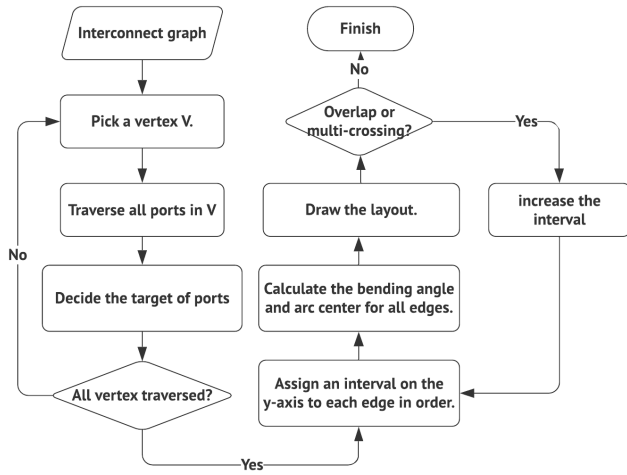


Fig. 1. Flowchart for the routing process.

top or bottom edge, the arc centers and the angle of the bends. Fig. 1 shows the high-level description of the routing process.

In a general formulation, above the routing area there are M PMTs named from V_1 to V_m from left to right, and below the routing area there are N PMTs named from V_{m+1} to V_{m+n} from left to right. For each PMT there are W Ports named from p_1 to p_w from left to right.

Figure 2 illustrates the topology for the algorithm. The algorithm starts with determining the connection relationships by finding the target PMT V_e for each port in each PMT. For a port p_i in a PMT V_i above the area (i in $[1, M]$), the algorithm traverses other PMTs in an order: $V_{i-1}, \dots, V_1, V_{m+1}, \dots, V_{m+n}, V_{i+1}, \dots, V_m$. Once an PMT V_e is found connected to V_i with an unarchived edge E_i in the graph, the p_i is marked with V_e , indicating that p_i should be routed to V_e , and then the edge E_i is archived for V_i to avoid duplication.

Then the algorithm traverses the right end pi of channels from left to right, to decide the distance y_i , where y_i is y_{i-1} plus a Δy . Here, y_0 is the bending radius plus a gap, and the

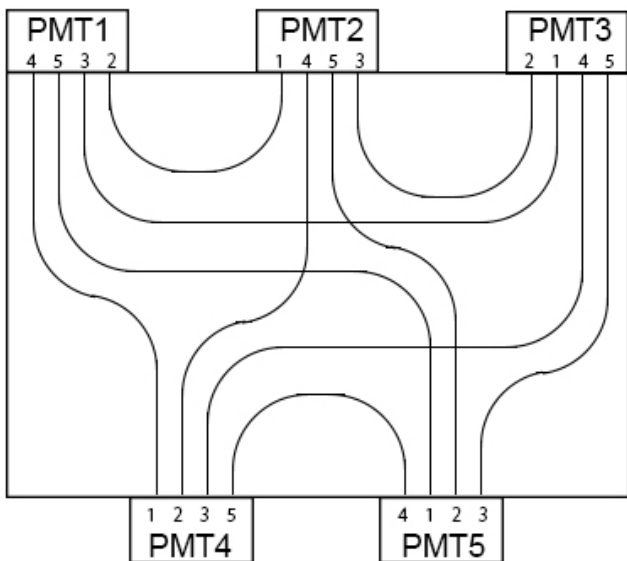


Fig. 2. An example of layout topology with 10 channels. The upper and lower PMT are initially spaced alternately to ensure the pitch. Each PMT has W ports and each port is marked with its target PMT.

Δy is set to a pitch between waveguides. Similar operations are performed for PMTs on the bottom. These determine the two endpoints and the distance of the horizontal line from the top or bottom edges. With a proper bending radius, the bending angle and center of each channel can be simply calculated so that all channels are decided.

For a high-density polymer waveguide layout, where the spacing between neighboring PMTs is below the bending radius, it is likely that there are multiple crossings, which disobey the principles we proposed above. To eliminate the multiple crossings, two ways can be considered. One is to increase the Δy can have the similar effect on eliminating the multiple crossing as increasing the spacing between neighboring PMTs. This way reaches its limit when the routing area is not broad enough. Another way is to reduce the bending radii so that the spacing is relatively large.

Figure 3 shows a routing result with 512 channels and a waveguide bending radius of 4 mm. The layout follows the three principles and makes full use of the area. Most crossing angles are 90 degrees, ensuring lower losses.

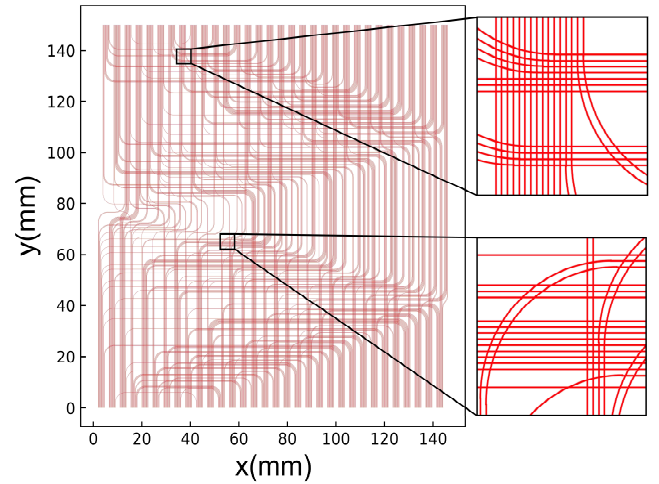


Fig. 3. 512 channel automatic waveguides layout within an area of $150 \times 150 \text{ mm}^2$.

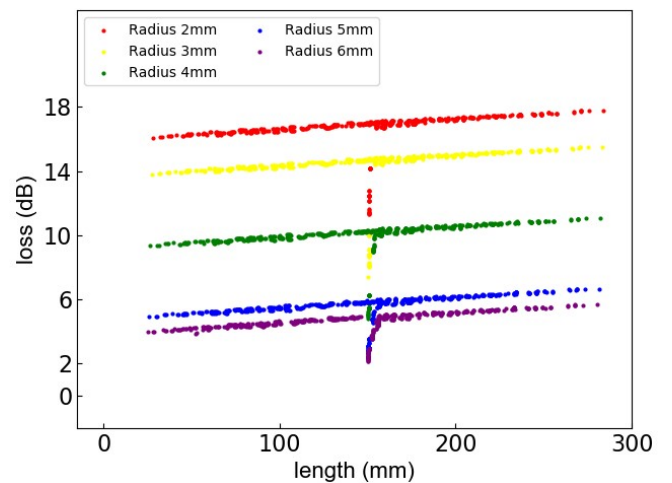


Fig. 4. The loss distribution of waveguides with different bending radii versus the waveguide length.

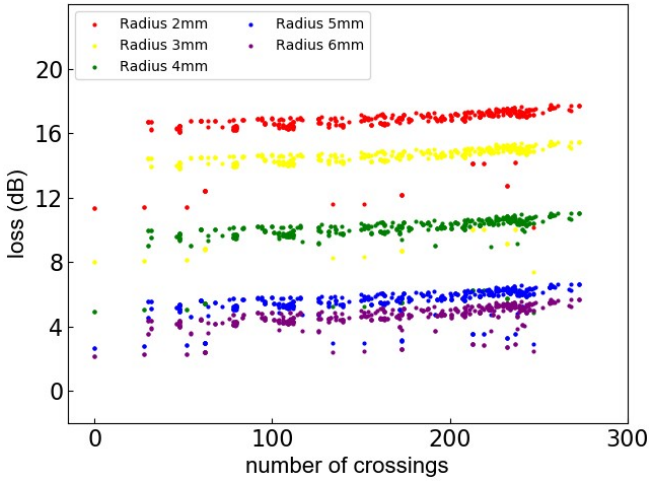


Fig. 5. The loss distribution of waveguides with different bending radii versus the number of crossings.

In this paper, we adopted a 0.05 dB/cm loss for straight waveguides which were measured results from waveguides fabricated by a standard photolithography method using commercially available UV-curable resins (core: NTT-AT E3135 and clad: NTT-AT E3129) as core and clad materials. We measured the crossover loss of polymer optical waveguides with 90-degree crossings. The measured crossover loss at 90 degrees is about 0.018 dB. Simulation results show that the loss at a small angle will be greater but as long as the angle exceeds 20 degrees, the loss will not exceed 0.01 dB. We also measured the loss of 90-degree bending structure in different radii as listed in Table 1, and the results are comparable to results reported in [10].

It can be observed from Fig. 4 that the loss is essentially linear with the waveguide length, but the loss of a waveguide with a length of about 150 mm deviates largely from this linear relationship, due to the absence of a bending structure. Fig.5 shows the relations between loss and number of crossings. There is a positive correlation between the loss and the number of crossings. Since the longer the waveguide is, the more likely it is to intersect with other waveguides, and the greater the waveguide loss is. The effects of waveguide length and number of crossing are synchronous.

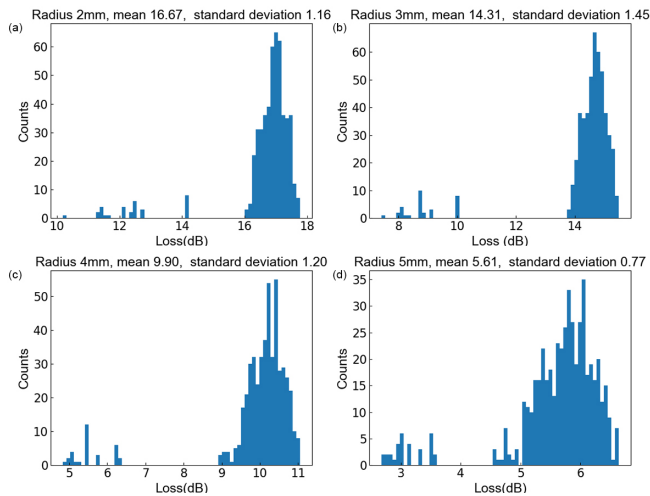


Fig. 6 The loss distribution with different bending radii.

Figure 6 shows the loss distribution of waveguides under different bending radii. The smaller the bending radius is, the greater the mean loss and standard deviation are. As the TABLE II showed, even when the radius is 5 mm, a 90-degrees bending loss is much greater than the straight waveguide loss and cross loss. This explains that the waveguide loss is significantly smaller when its length is around 150 mm, who has a small bending angle so the loss caused by the bending is much smaller.

In order to reduce overall loss, it is most effective to adopt a larger bend. However, using large bends often leads to crossover of multiple waveguides, which is contrary to the principles. In our case, a few multi-crosses appear with a radius of 5 mm while no multi-cross when the radius is 4 mm. As a result, there exists a trade-off between the waveguide loss and the size of the routing area and the total number of channels. Additional optimization algorithms may be further adopted to eliminate multi-cross and thus alleviate this trade-off.

III. CONCLUSION

In conclusion, we proposed a layout method for high-density on-board optical waveguide interconnection and the principles for optical routing works. We demonstrated a layout with 512 channels which prove the feasibility of this method, and compared the routing quality at different radii, where the estimated average channel loss is 9.90 dB and 5.61 dB when the radius is 4 mm and 5 mm, respectively.

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